Chapter 7

Discrete random variables and the binomial distribution

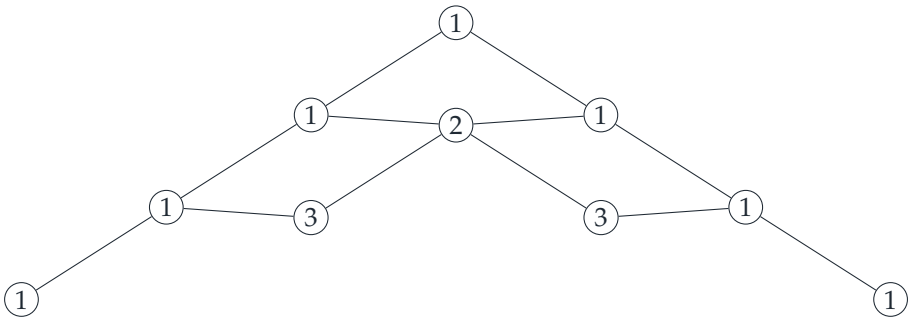
Skills: define a distribution, calculate and interpret an expected value, recognise and use a binomial distribution

7.1 Reference notes

A discrete random variable X assigns a real number to each outcome. Its distribution gives the values x_i and the probabilities $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

Key points

The expected value is $E(X) = \sum_i x_i p_i$. It represents the long-run average value when the experiment is repeated under the same conditions; it is not necessarily a value actually taken by X.

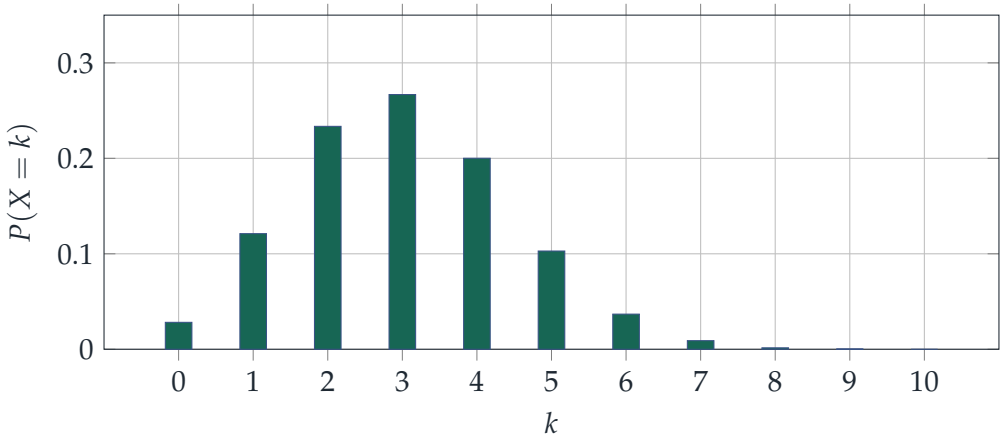


Key points

X follows a binomial distribution $\mathcal{B}(n, p)$ when X counts the number of successes in n independent Bernoulli trials with the same probability of success p. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad E(X) = np.$$

The binomial coefficient $\binom{n}{k}$ counts the number of paths containing exactly k successes among n trials. For $n \leq 10$, it can be read from Pascal's triangle.



The diagram approximately represents $\mathcal{B}(10, 0.3)$; its centre is close to $E(X) = 3$.

Session 47 — Expected value of a random variable

Discrete random variables

6 Apr 2027 • B: 2 h – C: 1 h

Session objectives

- Define a distribution.
- Calculate an expected value.
- Interpret a theoretical average.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 8 - x + 2$. .....
2. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. .....
3. Calculate $\binom{5}{2}$. .....
4. Calculate the expected value of the constant X equal to 8. .....
5. Express $X \geq 1$ using a complement. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: define a distribution and calculate an expected value.

Investigation

A game costs 2 euros. The gross winnings are 0, 3 or 10 euros with probabilities 0.6, 0.3, 0.1. Construct the distribution of the net gain X, calculate $E(X)$ and decide whether the game is favourable to the player.

Written record

The distribution of X assigns to each value x_i the probability p_i. The expected value $E(X) = \sum x_i p_i$ is a theoretical long-run average.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Practice – essential

1. Justify that the number of successes among 8 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Textbook references

Further practice: Calao, course pp. 154–159; exercises p. 160, no. 1–5. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application exercise

To launch a new perfume, a cosmetics chain sells its customers a scratch card for €3. The cards are printed in booklets of 25: in each booklet, 16 cards are losing cards, 6 cards give a €6 shopping voucher and 3 cards give a €13 shopping voucher. A customer buys a card chosen at random from a new booklet.
Let G be the random variable equal to the customer's net gain, in euros, that is, the value of the voucher obtained minus the price of the card (a negative gain is a loss).

1. What values does G take?
2. Copy and complete the following table, which gives the probability distribution of G.

Value g_i of G			
$P(G = g_i)$			

3. Calculate the expected value $E(G)$. What can be said from the customer's point of view? And from the chain's point of view?
4. The chain wants to earn €0.40 on average per card sold, without changing the composition of the booklets. At what price should it sell the card?

Homework

Redo practice exercise 2 with the following data: replace the initial value by 355 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Session 48 — Binomial distribution: recognising a situation

Discrete random variables

8 Apr 2027 • B: 1 h – C: 2 h

Session objectives

- Recognise a Bernoulli scheme.
- Identify n and p.
- Define X.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 5 % of 180. .....
2. Calculate $\binom{5}{2}$. .....
3. Calculate the expected value of the constant X equal to 9. .....
4. Express $X \geq 1$ using a complement. .....
5. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: recognise a Bernoulli scheme and identify n and p.

Investigation

A customer accepts an offer with probability 0.25. An adviser contacts eight customers independently. Define X, justify the binomial distribution and identify n and p.

Written record

$X \sim \mathcal{B}(n, p)$ if X counts the successes in n independent trials with the same probability p.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Practice – essential

1. Justify that the number of successes among 9 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

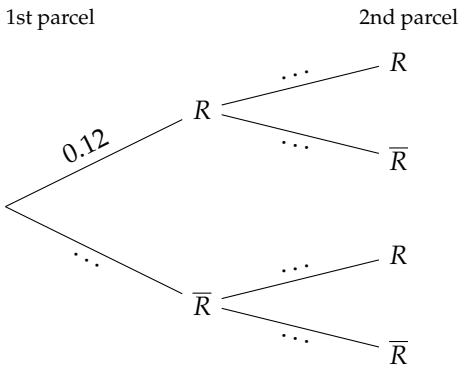
Textbook references

Further practice: Calao, course pp. 154–159; exercises p. 160, no. 5, then p. 162, no. 14–20. The application exercise below is original; these references point to textbook exercises for extra practice

and for its figures.

Application exercise

On a large online sales platform, 12 % of shipped parcels are returned by the customer. The quality department chooses two shipped parcels at random. The number of shipped parcels is large enough for this choice to be treated as two random draws with replacement.
Let R be the event "the parcel is returned" and X the random variable equal to the number of returned parcels among the two chosen parcels.



1. Copy and complete the weighted tree above.
2. Justify that X follows a binomial distribution and give its parameters.
3. Calculate $P(X = 0)$, $P(X = 1)$ and $P(X = 2)$. For $P(X = 1)$, state how many paths of the tree give the event $\{X = 1\}$.
4. Deduce, in two different ways, the probability that at least one of the two parcels is returned.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 360 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Session 49 — Pascal's triangle and binomial coefficients

Discrete random variables

27 Apr 2027 • B: 2 h – C: 1 h

Session objectives

- Build Pascal's triangle.
- Read $\binom{n}{k}$.
- Interpret a coefficient.

Fluency – 6 to 8 minutes, no calculator

1. Write the multiplier for an increase of 10 %. .....
2. Calculate the expected value of the constant X equal to 3. .....
3. Express $X \geq 1$ using a complement. .....
4. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. .....
5. Calculate $\binom{5}{2}$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: build Pascal's triangle and read $\binom{n}{k}$.

Investigation

Build rows $n = 0$ to $n = 6$ of Pascal's triangle. Explain the calculation of the number 15 in row 6, then read $\binom{6}{2}$ and $\binom{6}{4}$.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

Written record

Pascal's triangle satisfies $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. The edges equal 1.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Practice – essential

1. Justify that the number of successes among 10 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.

3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Textbook references

Further practice: Calao, p. 164, no. 24–29, then p. 166, no. 36. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application exercise

An online shop is testing two layouts of its home page (A/B test): for each new visitor, the site displays version A or version B at random, with the same probability $\frac{1}{2}$, independently of the other visitors. The next 7 visitors are observed and X denotes the random variable equal to the number of visitors who are shown version B.

1. Justify that X follows a binomial distribution and give its parameters.
2. Build Pascal's triangle up to row $n = 7$.
3. Read $\binom{7}{3}$ in the triangle and interpret this number in context. Calculate $P(X = 3)$ as a fraction, then rounded to 10^{-3}. Interpret the result.
4. Calculate the probability that at least 5 of the 7 visitors see version B (as a fraction, then rounded to 10^{-3}).

Homework

Redo practice exercise 2 with the following data: replace the initial value by 365 and the rate or common ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Session 50 — Binomial probabilities

Discrete random variables

29 Apr 2027 • B: 1 h – C: 2 h

Session objectives

- Calculate $P(X = k)$.
- Translate at least/at most.
- Use the complement.

Fluency – 6 to 8 minutes, no calculator

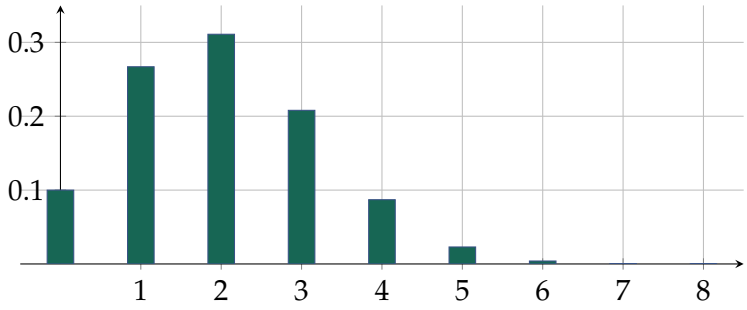
1. Simplify $3x + 4 - x + 2$. .....
2. Express $X \geq 1$ using a complement. .....
3. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. .....
4. Calculate $\binom{5}{2}$. .....
5. Calculate the expected value of the constant X equal to 4. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: calculate $p(x = k)$ and translate at least/at most.

Investigation

For $X \sim \mathcal{B}(8, 0.25)$, calculate $P(X = 2)$, $P(X = 0)$, then $P(X \geq 1)$. Translate each event into a sentence.



Written record

If $X \sim \mathcal{B}(n, p)$, then $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. Translate "at least", "at most" and "none" carefully.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Practice – essential

1. Justify that the number of successes among 6 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Textbook references

Further practice: Calao, p. 164, no. 24–29, then p. 168, no. 41–45. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application exercise

In a medical practice, it is estimated that a patient who booked an appointment online actually turns up for the consultation with probability 0.9. One morning, a doctor has 8 appointments booked online. Let X be the random variable equal to the number of patients who turn up among these 8. Assume that X follows a binomial distribution.
Probabilities will be rounded to 10^{-3}.

1. Give the parameters n and p of the distribution of X.
2. Calculate the probability of the following events: A: "all 8 patients turn up"; B: "at least one patient does not turn up".
3. Build Pascal's triangle up to row $n = 8$. Deduce $\binom{8}{6}$, then the probability of the event C: "exactly 6 patients turn up".
4. Express the event $\{X = 7\}$ in a sentence, then calculate $P(X = 7)$.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 370 and the rate or common ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Session 51 — Chapter review and test

Discrete random variables

4 May 2027 • B: 2 h – C: 1 h

Session objectives

- Calculate and interpret $E(X)$.
- Solve a binomial problem.
- Analyse one's mistakes.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 20 % of 100. .....
2. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. .....
3. Calculate $\binom{5}{2}$. .....
4. Calculate the expected value of the constant X equal to 5. .....
5. Express $X \geq 1$ using a complement. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: calculate and interpret $e(x)$ and solve a binomial problem.

Investigation

A centre receives 120 groups of 10 customers, each customer complaining with probability 0.08. Estimate the average number of complaints per group, then over all the groups. Discuss what the expected value does and does not allow us to predict.

Written record

For $X \sim \mathcal{B}(n, p)$, $E(X) = np$. The value gives the average number of successes over a large number of series of n trials.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Practice – essential

1. Justify that the number of successes among 7 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Textbook references

Further practice: Calao, pp. 160–169. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application exercise

A vegetable seed producer claims that 95 % of its tomato seeds germinate. The seeds are sold in packets of 12. Production is large enough for filling a packet to be treated as a random draw with replacement of 12 seeds. Let X be the random variable that associates with each packet the number of seeds that germinate.

1. What distribution does X follow? Justify and give its parameters.
2. Give $\binom{12}{11}$ and $\binom{12}{12}$.
3. Calculate $P(X = 11)$ and $P(X = 12)$, rounded to 10^{-3}. Interpret each of these results.
4. Deduce the probability, rounded to 10^{-3}, that a packet contains at least 11 seeds that germinate.
5. Calculate $E(X)$ and interpret this number for a gardener who regularly buys these packets.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 375 and the rate or common ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.